FIVE CHARACTERISATIONS OF ADEQUATE MODULES, LOCAL AND GLOBAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Adequacy is local and can be tested after restriction to affine schemes. Same chapter, Lemmas lemma-adequate-local and lemma-adequate-affine, invoked at proof lines 1816-1819. Gives the equivalence of (1) and (5).
- Adequacy is preserved by pullback along an arbitrary morphism. Same chapter, Lemma lemma-pullback-adequate, invoked at proof lines 1830-1832. Gives one direction of the equivalence of (4) with (1).

Source reference. Original lemma 06ZJ: sources/stacks-adequate/pinned-original-source.tex, source lines 1738–1738. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VI: sources/stacks-adequate/pinned-original-source.tex, source lines 1707–1707. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VJ: sources/stacks-adequate/pinned-original-source.tex, source lines 1759–1759. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be an $\mathcal{O}$-module on $(Sch/S)_\tau$. The following are equivalent*

(1) *$\mathcal{F}$ is adequate,*
(2) *there exists an affine open covering $S = \bigcup S_i$ and maps of quasi-coherent $\mathcal{O}_{S_i}$-modules $\mathcal{G}_i \to \mathcal{H}_i$ such that $\mathcal{F}|_{(Sch/S_i)_\tau}$ is the kernel of $\mathcal{G}_i^a \to \mathcal{H}_i^a$*
(3) *there exists a τ-covering $\{S_i \to S\}_{i\in I}$ and maps of $\mathcal{O}_{S_i}$-quasi-coherent modules $\mathcal{G}_i \to \mathcal{H}_i$ such that $\mathcal{F}|_{(Sch/S_i)_\tau}$ is the kernel of $\mathcal{G}_i^a \to \mathcal{H}_i^a$,*
(4) *there exists a τ-covering $\{f_i : S_i \to S\}_{i\in I}$ such that each $f_i^*\mathcal{F}$ is adequate,*

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(5) *for any affine scheme U over S the restriction $\mathcal{F}|_{(Sch/U)_\tau}$ is the kernel of a map $\mathcal{G}^a \to \mathcal{H}^a$ of quasi-coherent $\mathcal{O}_U$-modules.*

Proof. Let $U = \mathrm{Spec}(A)$ be an affine scheme over S. Set $F = F_{\mathcal{F},A}$. By definition, the functor F is adequate if and only if there exists a map of A-modules $M \to N$ such that $F = \mathrm{Ker}(\underline{M} \to \underline{N})$. Combining with Lemmas 06VI and 06ZJ we see that (1) and (5) are equivalent.

It is clear that (5) implies (2) and (2) implies (3). If (3) holds then we can refine the covering $\{S_i \to S\}$ such that each $S_i = \mathrm{Spec}(A_i)$ is affine. Then we see, by the preliminary remarks of the proof, that $F_{\mathcal{F},A_i}$ is adequate. Thus $\mathcal{F}$ is adequate by definition. Hence (3) implies (1).

Finally, (4) is equivalent to (1) using Lemma 06VJ for one direction and that a composition of τ-coverings is a τ-covering for the other. □